Lebanese University
Faculty of Sciences
Section V

6/11/2025
Duration: 90 mins

Applied Mathematics and Statistics for Computational Sciences

Problem 1 : Principal Component Analysis (PCA)

A dataset contains $n = 200$ observations described by $p = 4$ centered and standardized variables (X_1, X_2, X_3, X_4). A Principal Component Analysis (PCA) is performed on this dataset.

The eigenvalues of the empirical covariance matrix are given in the table below:

Component	PC1	PC2	PC3	PC4
Eigenvalue	2.60	0.90	0.40	0.10

The loadings of the first two principal components are:

$$\text{PC1} = 0.55X_1 + 0.52X_2 + 0.50X_3 + 0.42X_4$$
$$\text{PC2} = -0.10X_1 - 0.20X_2 + 0.75X_3 - 0.62X_4$$

1) Compute the percentage of total variance explained by PC1, PC2, PC3 and PC4. Based on these results, how many components would you retain for dimension reduction? Justify your answer.

2) Interpret the first principal component PC1 in terms of the original variables. What type of information does this component summarize?

3) Interpret the second principal component PC2. Explain how its interpretation differs from that of PC1.

4) Suppose the data are projected onto the first two principal components. Describe the dimension of the transformed dataset, explain what type of information is lost by discarding the third and fourth principal components, and discuss whether this projection is suitable for visualization purposes.

5) Which of the following statements about PCA is **correct**?

(A) PCA finds orthogonal directions that maximize the variance of the projected data.

(B) PCA preserves class labels during dimension reduction.

(C) PCA minimizes the reconstruction error under the ℓ_1 norm.

(D) PCA requires the data to follow a normal distribution.

Problem 2 : Optimization and Gradient Descent

Consider the function

$$f(x) = (x-3)^2.$$

1) Compute the gradient of the function f.

2) Starting from the initial point $x^{(0)} = 0$ and using a learning rate $\eta = 0.2$, apply the gradient descent algorithm to minimize $f(x) = (x-3)^2$ and compute the iterates until the method has practically converged. You must justify your stopping point by a clear analysis.

3) Write the pseudocode or a Python-like algorithm implementing gradient descent for minimizing a one-dimensional differentiable function.

Now consider the empirical risk function

$$F(\theta) = \frac{1}{N}\sum_{i=1}^{N}(\theta - y_i)^2,$$

where N is large.

4) Explain the difference between full-batch gradient descent, mini-batch gradient descent, and stochastic gradient descent, focusing on how the gradient is computed at each iteration.

5) Which of the following statements about gradient descent methods is **correct**?

(A) Increasing the learning rate always accelerates convergence.

(B) Gradient descent always converges to the global minimum, regardless of the function.

(C) Mini-batch gradient descent provides a trade-off between convergence stability and computational efficiency.

(D) Stochastic gradient descent requires computing the gradient using the entire dataset.